

Neutron Attenuation

NE 630

Lesson 04

Prepare: Read FNRP Sections 2.1–2.2. Define the key terms and distinguish material attenuation from geometric attenuation before class.

Macro objective

Characterize the attenuation of neutrons in a parallel beam or emitted from a point source as a function of distance and material composition.

Vocabulary: transport; target number density n_X ; beam intensity I ; microscopic cross section σ ; barn; macroscopic cross section Σ ; survival fraction; first collision; mean free path; uncollided flux ϕ_u ; point source; material/geometric attenuation; atom enrichment (a/o); mass/weight enrichment (w/o)

1 What moves, and what attenuates?

Idealized neutron-transport picture

- Neutrons are neutral.
- Neutrons travel in straight lines between interactions.
- Upon exiting an interaction, a neutron's direction and energy are different.
- Neutronic systems have enough neutrons such that **averages** are useful.
- The neutron (number) density in these systems is much smaller than that of the surrounding medium.
- The surrounding medium is **isotropic**.

For a one-direction beam,

$$I(x) = v n_u'''(x) = \phi_u(x), \quad [I] = \text{n cm}^{-2} \text{s}^{-1}.$$

Attenuation is not necessarily absorption

A neutron scatters without being absorbed. Does it still contribute to (a) the original beam and (b) the total neutron population?

(a) (b)

2 From material density to target density

Number-density conversion

$$n_X = \frac{\rho_X N_A}{M_X}, \quad N_A = 6.022 \times 10^{23} \text{ entities/mol.}$$

Here n_X means *target* atoms or molecules per volume (Lewis writes N), not neutron density. Keep the species label in the units.

Examples

Al: $\rho = 2.70 \text{ g/cm}^3$, $M = 26.98 \text{ g/mol}$,

$$n_{\text{Al}} = \text{..... Al atoms/cm}^3.$$

Water: $\rho = 1.00 \text{ g/cm}^3$, $M = 18.015 \text{ g/mol}$,

$$n_{\text{H}_2\text{O}} = \text{..... molecules/cm}^3,$$

$$n_{\text{H}} = 2n_{\text{H}_2\text{O}} = \text{..... H atoms/cm}^3.$$

Why is stoichiometry applied after counting molecules?

3 From a nuclear area to a material coefficient

Count what a thin slice removes

Let $A = \Delta_y \Delta_z$ be the beam area, Δx the slice thickness, n the target-nucleus density, and σ the area presented by one nucleus. The slice contains

$$N_{\text{slice}} = n A \Delta x.$$

Complete the blocked-area fraction:

$$\frac{N_{\text{slice}} \sigma}{A} = \text{.....} \equiv \Sigma \Delta x, \quad \Sigma = \text{.....}$$

For a thin slice, the balance on uncollided intensity is

$$I(x + \Delta x) \approx I(x)(1 - \Sigma \Delta x).$$

Subtract $I(x)$, divide by Δx , and take $\Delta x \rightarrow 0$:

$$\frac{dI}{dx} = \text{.....} \Rightarrow I(x) = \text{.....}$$

Microscopic versus macroscopic

quantity	meaning	units
σ	effective area of one target nucleus	$\text{cm}^2/\text{nucleus or b}$
$\Sigma = n\sigma$	interaction probability per unit path length in a material	cm^{-1}
Σx	dimensionless path length through the material	1

1 b = 10^{-24} cm^2 . For attenuation, $\sigma = \sigma_t = \sigma_s + \sigma_a$ and $\Sigma = \Sigma_t$: any collision counts. At fixed nuclide and neutron energy, changing density changes, not

The Lesson 3 analogy

	radioactive decay	attenuation
independent variable	t	x
uncollided/undecayed amount	$N(t)$	$I(x)$
removal coefficient	$\lambda [\text{s}^{-1}]$	$\Sigma [\text{cm}^{-1}]$
characteristic scale	$1/\lambda$	$1/\Sigma$

Infer a cross section from transmission

A thermal-neutron beam crosses 0.050 cm of aluminum and emerges with $I/I_0 = 0.9949$. Using n_{Al} from Section 2,

$$\Sigma = -\frac{\ln(I/I_0)}{x} = \text{.....},$$

$$\sigma = \frac{\Sigma}{n_{\text{Al}}} = \text{.....}$$

Why is the measured loss fraction not itself a cross section?

4 Read attenuation as a probability law

First-flight distribution in a uniform material

Let X be the distance to a neutron's first collision. Then

$$\underbrace{P(X > x)}_{\text{survive to } x} = e^{-\Sigma x}, \quad \underbrace{p_X(x)}_{\text{first-collision density}} = \Sigma e^{-\Sigma x},$$

$$P(X \leq x) = 1 - e^{-\Sigma x}.$$

The mean free path is

$$\ell \equiv E[X] = \int_0^\infty x \Sigma e^{-\Sigma x} dx = \frac{1}{\Sigma}.$$

Lewis uses λ for mean free path; ℓ avoids confusion with the Lesson 3 decay constant.

One mean free path

At $x = \ell$, the fraction still uncollided is

$$P(X > \ell) = \underline{\hspace{2cm}} = \dots\dots\dots,$$

and the fraction having collided is

$$P(X \leq \ell) = \underline{\hspace{2cm}} = \dots\dots\dots$$

Is ℓ the distance by which one-half survive? Explain.

Probability-density checks

$$\int_0^\infty p_X(x) dx = \underline{\hspace{2cm}}, \quad [p_X] = \underline{\hspace{2cm}}.$$

Why does p_X carry units while $P(X > x)$ does not?

5 Parallel beam versus point source

Two uncollided-flux geometries

Collimated parallel beam:

$$\phi_u(x) = \phi_u(0) \underbrace{e^{-\Sigma x}}_{\text{material}}.$$

Isotropic point source of strength s_p [n/s]:

$$\phi_u(r) = s_p \underbrace{\frac{1}{4\pi r^2}}_{\text{geometric}} \underbrace{e^{-\Sigma r}}_{\text{material}}.$$

Geometry spreads surviving neutrons over a larger area; material attenuation removes neutrons from their uncollided first flight. For $r_2 = 2r_1$ in the same medium,

$$\frac{\phi_u(2r_1)}{\phi_u(r_1)} = \underline{\hspace{2cm}}.$$

In vacuum this becomes $\dots\dots\dots$. Which attenuation remains, and why does a parallel beam not have this factor?

6 Build a material from its composition

Three reusable composition routes

For a compound with molecular number density n_m and v_i atoms of species i per molecule,

$$n_m = \frac{\rho N_A}{M_m}, \quad n_i = v_i n_m, \quad \boxed{\Sigma = \sum_i n_i \sigma_i}.$$

For isotope atom fractions $x_i = n_i / (\sum_j n_j)$,

$$\bar{M} = \sum_i x_i M_i, \quad n_i = x_i \frac{\rho N_A}{\bar{M}}.$$

For isotope mass fractions $w_i = m_i / (\sum_j m_j)$,

$$\boxed{n_i = \frac{\rho w_i N_A}{M_i}}.$$

Use cross sections at the same neutron energy and for the same reaction category (total here). Atom and mass fractions enter through n_i ; volume fractions instead weight the constituents' macroscopic cross sections. For water, $n_H : n_O = \underline{\hspace{2cm}}$ and $\Sigma_{H_2O} = n_{H_2O} \underline{\hspace{2cm}}$.

Composition logic

If the water density decreases while its molecular composition stays fixed, which of n_i , σ_i , and Σ change?

7 Translate uranium enrichment

Atom and mass (weight) enrichment

For a ^{235}U – ^{238}U mixture,

$$e_a = \frac{n_{235}}{n_{235} + n_{238}}, \quad e_w = \frac{m_{235}}{m_{235} + m_{238}}.$$

Convert a stated mass enrichment to atom enrichment with

$$\boxed{e_a = \frac{e_w / M_{235}}{e_w / M_{235} + (1 - e_w) / M_{238}}}.$$

Uranium enriched to 4.00 wt% ^{235}U

Choose a 100 g uranium basis and neglect ^{234}U :

isotope	mass	amount (mol)
^{235}U	4.00 g	$\dots\dots\dots$
^{238}U	96.00 g	$\dots\dots\dots$

$$e_a = \underline{\hspace{2cm}} = \dots\dots\dots$$

Why is the atom percent slightly larger than the mass percent? $\dots\dots\dots$